

In Class Exercise

February 25, 2026

We are given an integer program defined as follows:

$$\begin{aligned} &\text{maximize} && 10x_1 + 22x_2 + 5x_3 + 15x_4 + 17x_5 + 12x_6 + 4x_7 \\ &\text{subject to} && 5x_1 + 3x_2 + 8x_3 + 9x_4 + 16x_5 + 5x_6 + 10x_7 \leq 600, \\ &&& x_i \in \{0, 1\} \quad (i = 1, 2, 3), \\ &&& 0 \leq x_i \leq 400 \quad (i = 4, 5, 6, 7). \end{aligned}$$

For each of the parts below, you are to add constraint(s) and possibly variables to ensure that the logical condition is satisfied by the integer program. Each part is independent; that is, no part depends on the parts preceding it. You do not need to repeat the integer programming objective or constraints given above. You may use the big M for formulating constraint when it is appropriate.

1. Write a single linear constraint equivalent to “If $x_2 = 1$ then $x_1 = 0$ ”.
2. Add three binary variables w_5, w_6, w_7 and exactly two constraints that ensure $x_5 \in \{9, 15, 20\}$.
3. Add variable(s) and constraint(s) that ensure either $2x_4 + x_5 \leq 50$ or $4x_4 - x_5 \geq 20$, but not both.
4. Add two or three binary variables and at most four constraints to ensure that at least two of the following hold: (i) $x_4 \geq 50$, (ii) $x_5 \leq 25$, (iii) $x_6 + x_7 \leq 100$.